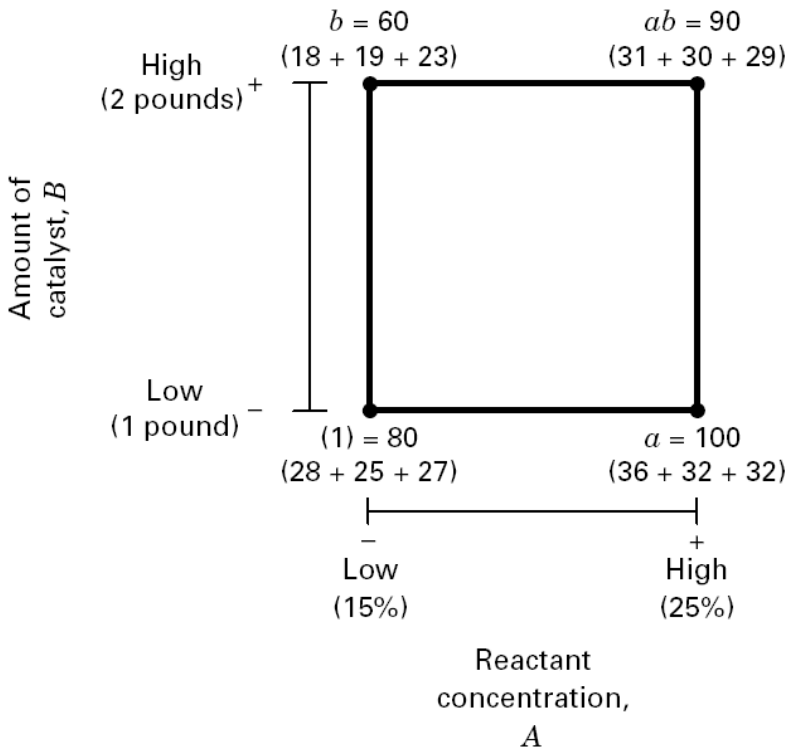
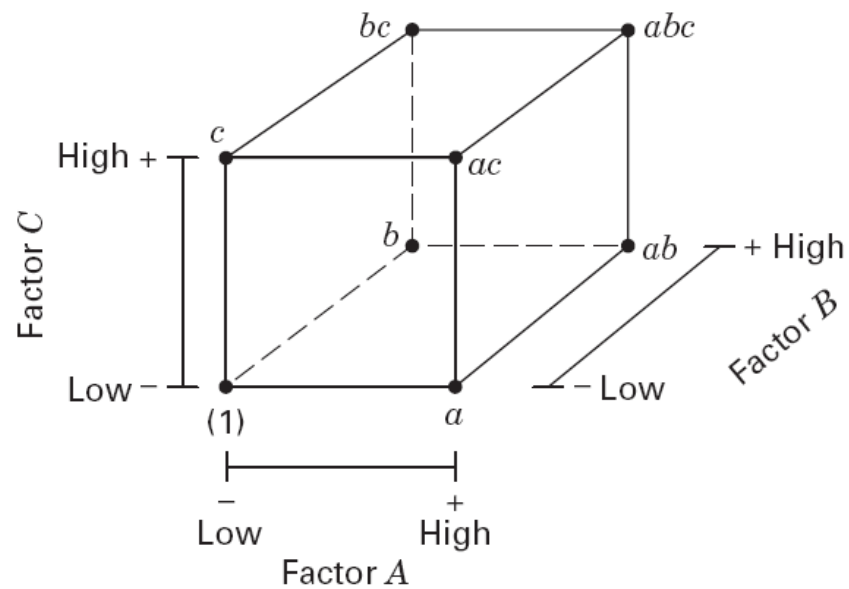
The 2^3 Factorial Design





(a) Geometric view

Effects in The 2^3 Factorial Design

$$A = \bar{y}_{A^+} - \bar{y}_{A^-} = \frac{a+ab+ac+abc}{4n} - \frac{(1)+b+c+bc}{4n}$$

$$= \frac{1}{4n}[a + ab + ac + abc - (1) - b - c - bc]$$

$$B = \bar{y}_{B^+} - \bar{y}_{B^-} = \frac{1}{4n}[b + ab + bc + abc - (1) - a - c - ac]$$

$$C = \bar{y}_{C^+} - \bar{y}_{C^-} = \frac{1}{4n}[c + ac + bc + abc - (1) - a - b - ab]$$

$$AB = \bar{y}_{(AB)+} - \bar{y}_{(AB)-} = \frac{1}{4\mathrm{n}}[abc + ab + c + (1) - bc - b - ac - a]$$

$$ABC = \bar{y}_{(ABC)+} - \bar{y}_{(ABC)-} = \frac{1}{4\mathrm{n}}[abc + a + b + c - ab - ac - bc - (1)]$$

Contrasts

Algebraic Signs for Calculating Effects in the 2^3 Design

	Factorial Effect							
Treatment Combination	I	A	B	AB	C	AC	BC	ABC
(1)	+	−	−	+	−	+	+	−
a	+	+	−	−	−	−	+	+
b	+	−	+	−	−	+	−	+
ab	+	+	+	+	−	−	−	−
c	+	−	−	+	+	−	−	+
ac	+	+	−	−	+	+	−	−
bc	+	−	+	−	+	−	+	−
abc	+	+	+	+	+	+	+	+

ABC is called 3-factor interaction effect

An Example of a 2^3 Factorial Design

The Plasma Etch Experiment, Example 6.1						
	Coded Factors			Etch Rate		
Run	A	B	C	Replicate 1	Replicate 2	Total
1	−1	−1	−1	550	604	(1) = 1154
2	1	−1	−1	669	650	a = 1319
3	−1	1	−1	633	601	b = 1234
4	1	1	−1	642	635	ab = 1277
5	−1	−1	1	1037	1052	c = 2089
6	1	−1	1	749	868	ac = 1617
7	−1	1	1	1075	1063	bc = 2138
8	1	1	1	729	860	abc = 1589

A = gap, B = Flow, C = Power, y = Etch Rate

Effects in The Example

$$A = \frac{1}{4n}[a-(1)+ab-b+ac-c+abc-bc] = \frac{1}{8}[-813] = -101.625$$

$$B = \frac{1}{4n}[b+ab+bc+abc-(1)-a-c-ac] = \frac{1}{8}[59] = 7.375$$

$$C = \frac{1}{4n}[c+ac+bc+abc-(1)-a-b-ab] = \frac{1}{8}[2449] = 306.125$$

$$AB = \frac{1}{4n}[ab-a-b+(1)+abc-bc-ac+c] = \frac{1}{8}[-199] = -24.875$$

$$AC = \frac{1}{4n}[(1)-a+b-ab-c+ac-bc+abc] = \frac{1}{8}[-1229] = -153.625$$

$$BC = \frac{1}{4n}[(1)+a-b-ab-c-ac+bc+abc] = \frac{1}{8}[-17] = -2.125$$

$$ABC = \frac{1}{4n}[abc-bc-ac+c-ab+b+a-(1)] = \frac{1}{8}[45] = 5.625$$

Sum of Squares

Recall that for a contrast C:

$$SS_C = \frac{\left(\sum_{i=1}^{2^k} c_i \bar{y}_i\right)^2}{\frac{1}{n}\sum_{i=1}^{2^k} c_i^2} = \frac{\left(\sum_{i=1}^{2^k} c_i y_{i\cdot}\right)^2}{n\sum_{i=1}^{2^k} c_i^2}$$

Here $\bar{y}_i$ is the average response of the ith treatment combination, and $y_{i\cdot} = n\bar{y}_i$ is the total.

For a 2^3 factorial design:

$$SS_C = \frac{\left(\sum_{i=1}^{2^3} c_i y_{i\cdot}\right)^2}{n\sum_{i=1}^{2^3} c_i^2} = \frac{\left(\sum_{i=1}^{2^3} c_i y_{i\cdot}\right)^2}{8n}$$

SS in The Example

$$SS_A = \frac{(-813)^2}{16} = 41{,}310.5625$$

$$SS_B = \frac{(59)^2}{16} = 217.5625$$

$$SS_C = \frac{(2449)^2}{16} = 374{,}850.0625$$

$$SS_{AB} = \frac{(-199)^2}{16} = 2475.0625$$

$$SS_{AC} = \frac{(-1229)^2}{16} = 94{,}402.5625$$

$$SS_{BC} = \frac{(-17)^2}{16} = 18.0625$$

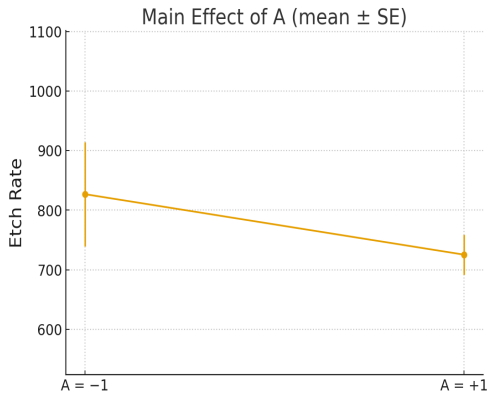
$$SS_{ABC} = \frac{(45)^2}{16} = 126.5625$$

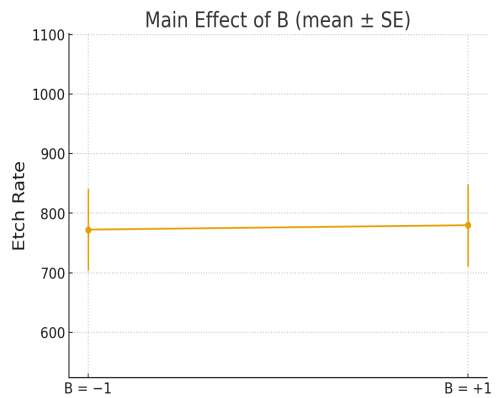
ANOVA Table

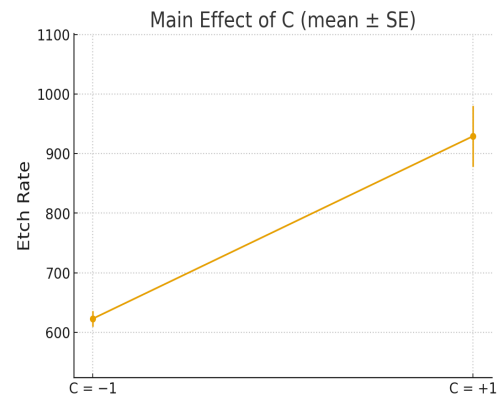
Analysis of Variance for the Plasma Etching Experiment					
Source of Variation	**Sum of Squares**	**Degrees of Freedom**	**Mean Square**	F_0	***P*-Value**
Gap (*A*)	41,310.5625	1	41,310.5625	18.34	0.0027
Gas flow (*B*)	217.5625	1	217.5625	0.10	0.7639
Power (*C*)	374,850.0625	1	374,850.0625	166.41	0.0001
AB	2475.0625	1	2475.0625	1.10	0.3252
AC	94,402.5625	1	94,402.5625	41.91	0.0002
BC	18.0625	1	18.0625	0.01	0.9308
ABC	126.5625	1	126.5625	0.06	0.8186
Error	18,020.5000	8	2252.5625		
Total	531,420.9375	15			

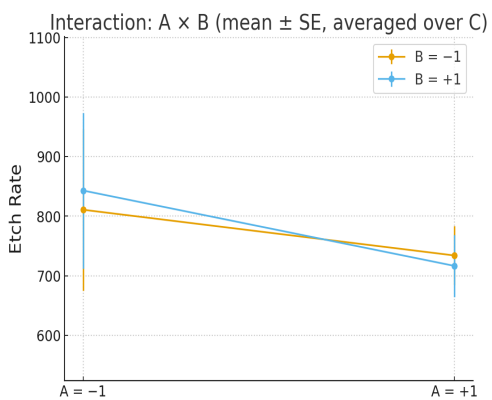
Question: what if n=1, that is, each treatment combination has a single replicate?

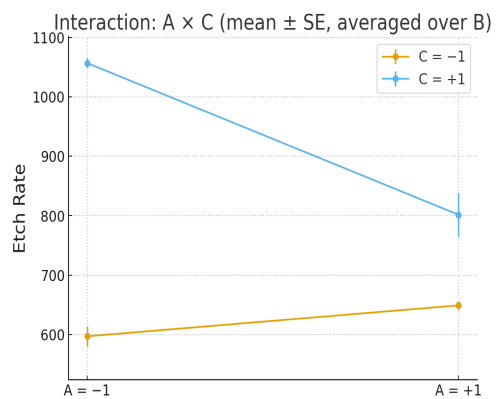
Main Effects & Two-Factor Interactions

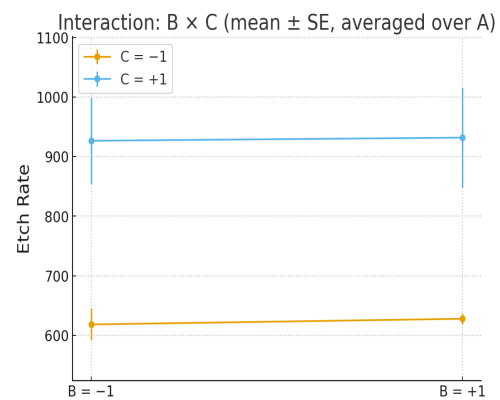












Synergy and Antagonism

- **General definition**: When two or more elements (e.g., drugs, genes, business strategies, or people) interact in a way that their combined effect is greater than the sum of their individual effects.
- **Example** : If Drug A reduces tumor size by 20% and Drug B by 30%, but together they reduce it by 70%, that's **synergy.**

- The opposite of synergy is **antagonism**. When two or more elements interact in a way that their combined effect is weaker than expected from their individual contributions.

- Visual rules
  - Parallel lines → no interaction (additive).
  - Lines fan apart as A goes from − to + (slope is steeper when C=+ than when C=−) → synergy (positive interaction).
  - Lines converge or cross (slope is smaller when C=+ than when C=−) → antagonism (negative interaction).

Regression and Response Surface

$$y = \beta_0 + \beta_1 x_1 + \beta_3 x_3 + \beta_{13} x_1 x_3 + \epsilon$$

Coded variables

Algebraic Signs for Calculating Effects in the 2^3 Design

	Factorial Effect							
Treatment Combination	***I***	***A***	***B***	***AB***	***C***	***AC***	***BC***	***ABC***
(1)	+	−	−	+	−	+	+	−
a	+	+	−	−	−	−	+	+
b	+	−	+	−	−	+	−	+
ab	+	+	+	+	−	−	−	−
c	+	−	−	+	+	−	−	+
ac	+	+	−	−	+	+	−	−
bc	+	−	+	−	+	−	+	−
abc	+	+	+	+	+	+	+	+

- The sum of the product of signs in any two columns is zero
- **Orthogonal** design
- Orthogonality is an important property shared by all factorial designs

Regression and Response Surface

$$\hat{\beta} = (X'X)^{-1}\, X'y$$

$$\begin{aligned}\hat{y} &= \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_3 x_3 + \hat{\beta}_{13} x_1 x_3 \\ &= 776.0625 + \left(\frac{-101.625}{2}\right) x_1 + \left(\frac{306.125}{2}\right) x_3 + \left(\frac{-153.625}{2}\right) x_1 x_3\end{aligned}$$

Each component in $\hat{\beta}$ is half of its corresponding effect.

The General 2^k Factorial Design

There are k main effects, and

$$\binom{k}{2} \text{ two-factor interactions}$$

$$\binom{k}{3} \text{ three-factor interactions}$$

$$\vdots$$

$$1 \; k-\text{factor interaction}$$

Each effect has 1 df

2^k Factorial Designs

Simple generalization of the $\mathbf{2^3}$ design

Contrast Constants for the 2^4 Design

	A	*B*	*AB*	*C*	*AC*	*BC*	*ABC*	*D*	*AD*	*BD*	*ABD*	*CD*	*ACD*	*BCD*	*ABCD*
(1)	−	−	+	−	+	+	−	−	+	+	−	+	−	−	+
a	+	−	−	−	−	+	+	−	−	+	+	+	+	−	−
b	−	+	−	−	+	−	+	−	+	−	+	+	−	+	−
ab	+	+	+	−	−	−	−	−	−	−	−	+	+	+	+
c	−	−	+	+	−	−	+	−	+	+	−	−	+	+	−
ac	+	−	−	+	+	−	−	−	−	+	+	−	−	+	+
bc	−	+	−	+	−	+	−	−	+	−	+	−	+	−	+
abc	+	+	+	+	+	+	+	−	−	−	−	−	−	−	−
d	−	−	+	−	+	+	−	+	−	−	+	−	+	+	−
ad	+	−	−	−	−	+	+	+	+	−	−	−	−	+	+
bd	−	+	−	−	+	−	+	+	−	+	−	−	+	−	+
abd	+	+	+	−	−	−	−	+	+	+	+	−	−	−	−
cd	−	−	+	+	−	−	+	+	−	−	+	+	−	−	+
acd	+	−	−	+	+	−	−	+	+	−	−	+	+	−	−
bcd	−	+	−	+	−	+	−	+	−	+	−	+	−	+	−
abcd	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+

If each treatment combination is replicated n times:

$$df_{total} = 2^k n - 1$$

$$df_E = 2^k n - 1 - (2^k - 1) = 2^k(n-1)$$

We can compute SS and show ANOVA.

Analysis of Variance for a 2^k Design		
Source of Variation	**Sum of Squares**	**Degrees of Freedom**
k **main effects**		
A	SS_A	1
B	SS_B	1
$\vdots$	$\vdots$	$\vdots$
K	SS_K	1
$\binom{k}{\mathbf{2}}$ **two-factor interactions**		
AB	SS_{AB}	1
AC	SS_{AC}	1
$\vdots$	$\vdots$	$\vdots$
JK	SS_{JK}	1
$\binom{k}{\mathbf{3}}$ **three-factor interactions**		
ABC	SS_{ABC}	1
ABD	SS_{ABD}	1
$\vdots$	$\vdots$	$\vdots$
IJK	SS_{IJK}	1
$\vdots$	$\vdots$	$\vdots$
$\binom{k}{k}$ k**-factor interaction**		
$ABC\cdots K$	$SS_{ABC\cdots K}$	1
Error	SS_E	$2^k(n-1)$
Total	SS_T	$n2^k-1$

Unreplicated 2^k Factorial Designs

- When k is big, 2^k is big, replicates may be costly.
- We may only be able to obtain **one observation** at each treatment combination
- An unreplicated 2^k factorial design is also sometimes called a "**single replicate**" of the 2^k
- These designs are very widely used because they save experimental runs
- No df for estimating the error.

Unreplicated 2^k Factorial Designs

- Lack of replication causes potential **problems** in statistical testing
  - Replication admits an estimate of "pure error"
  - With no replication, fitting the full model results in zero degrees of freedom for error
- Potential **solutions** to this problem
  - Solution 1: Pooling high-order interactions to estimate error

    Because of the effect **Hierarchy** principle: lower-order effects are more likely to be important than higher-order effects (Higher-order effects are more likely to be unimportant)
  - Solution 2: Normal probability plotting of effects

Solution 1:
Pool high-order interactions to estimate error:

Analysis of Variance for a 2^k Design		
Source of Variation	**Sum of Squares**	**Degrees of Freedom**
k **main effects**		
A	SS_A	1
B	SS_B	1
$\vdots$	$\vdots$	$\vdots$
K	SS_K	1
$\binom{k}{2}$ **two-factor interactions**		
AB	SS_{AB}	1
AC	SS_{AC}	1
$\vdots$	$\vdots$	$\vdots$
JK	SS_{JK}	1
$\binom{k}{3}$ **three-factor interactions**		
ABC	SS_{ABC}	1
ABD	SS_{ABD}	1
$\vdots$	$\vdots$	$\vdots$
IJK	SS_{IJK}	1
$\vdots$	$\vdots$	$\vdots$
$\binom{k}{k}$ k**-factor interaction**		
$ABC\cdots K$	$SS_{ABC\cdots K}$	1
Error	SS_E	$2^k(n-1)$
Total	SS_T	$n2^k-1$

Example of an Unreplicated 2^k Design

A 2^4 factorial was used to investigate the effects of four factors on the response

	Factor					
Run Number	*A*	*B*	*C*	*D*	Run Label	Filtration Rate (gal/h)
1	−	−	−	−	(1)	45
2	+	−	−	−	*a*	71
3	−	+	−	−	*b*	48
4	+	+	−	−	*ab*	65
5	−	−	+	−	*c*	68
6	+	−	+	−	*ac*	60
7	−	+	+	−	*bc*	80
8	+	+	+	−	*abc*	65
9	−	−	−	+	*d*	43
10	+	−	−	+	*ad*	100
11	−	+	−	+	*bd*	45
12	+	+	−	+	*abd*	104
13	−	−	+	+	*cd*	75
14	+	−	+	+	*acd*	86
15	−	+	+	+	*bcd*	70
16	+	+	+	+	*abcd*	96

ANOVA by Pooling Higher-order Interactions to the Error

Analysis of Variance for the Pilot Plant Filtration Rate Experiment

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0	P-Value
A					
B					
C					
D					
AB					
AC					
AD					
BC					
BD					
CD					
Error					
Total					

Solution 2: Half-normal Plot

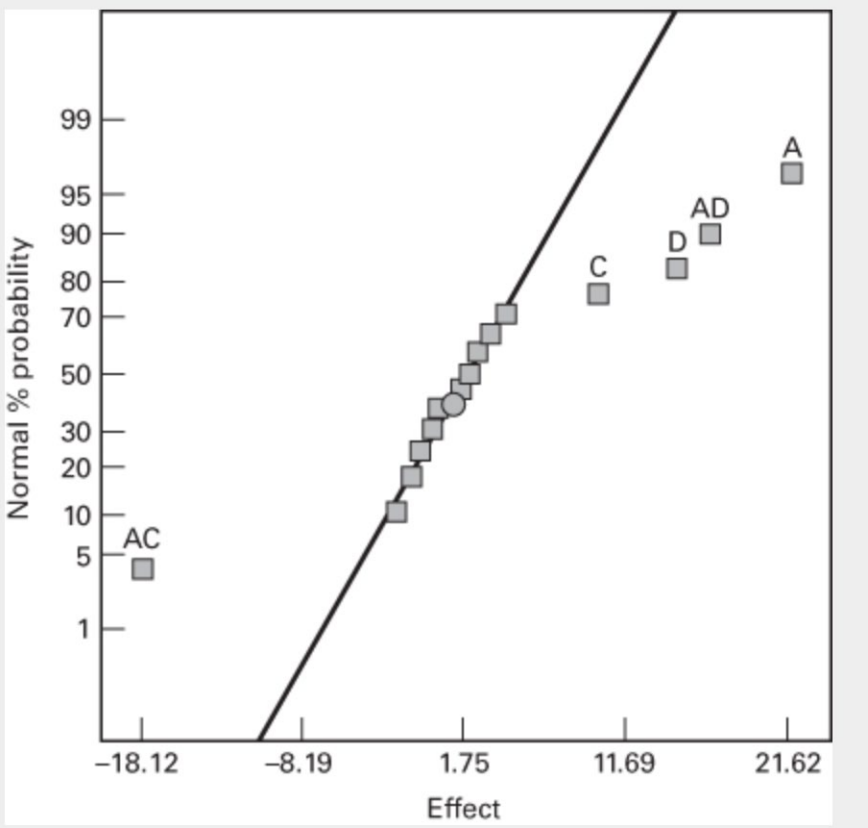
■ TABLE 6.12
Factor Effect Estimates and Sums of Squares for the 2^4 Factorial in Example 6.2

Model Term	Effect Estimate	Sum of Squares	Percent Contribution
A	21.625	1870.56	32.6397
B	3.125	39.0625	0.681608
C	9.875	390.062	6.80626
D	14.625	855.563	14.9288
AB	0.125	0.0625	0.00109057
AC	−18.125	1314.06	22.9293
AD	16.625	1105.56	19.2911
BC	2.375	22.5625	0.393696
BD	−0.375	0.5625	0.00981515
CD	−1.125	5.0625	0.0883363
ABC	1.875	14.0625	0.245379
ABD	4.125	68.0625	1.18763
ACD	−1.625	10.5625	0.184307
BCD	−2.625	27.5625	0.480942
ABCD	1.375	7.5625	0.131959

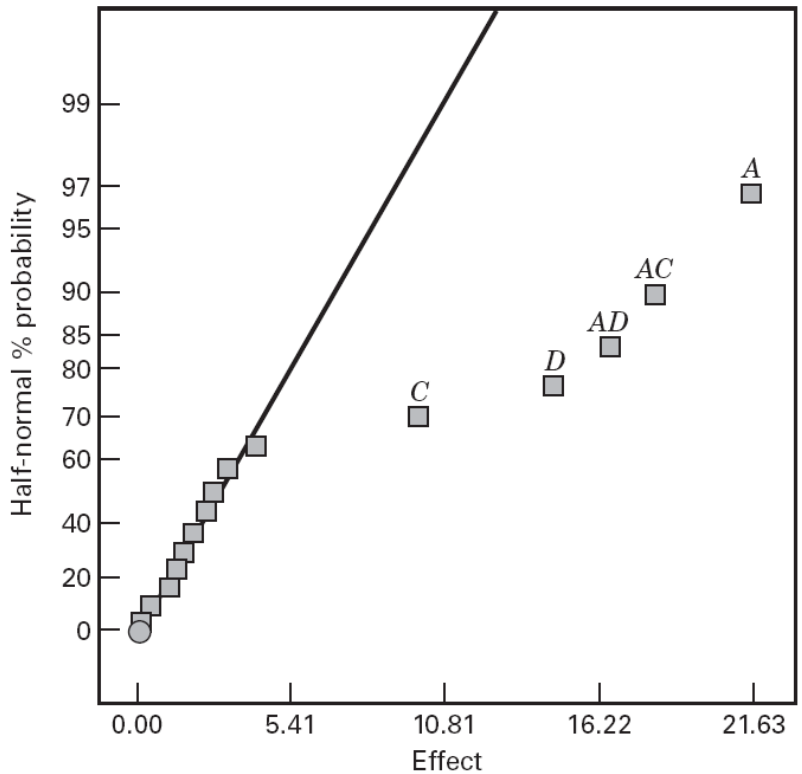
Estimates of the Effects

The estimated effects follow a Normal distribution if they are not significant!

Normal Probability Plot of Effects



The Half-Normal Probability Plot of Effects



R code: `library(DoE.base)`
`halfnormal(Effect, labs=Term)`
https://rdrr.io/cran/DoE.base/man/halfnormal.html

ANOVA Table

Analysis of Variance for the Pilot Plant Filtration Rate Experiment in *A*, *C*, and *D*

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0	*P*-Value
A	1870.56	1	1870.56		<0.0001
C	390.06	1	390.06		<0.0001
D	855.56	1	855.56		<0.0001
AC	1314.06	1	1314.06		<0.0001
AD	1105.56	1	1105.56		<0.0001
Error	195.14	10	19.51		
Total	5730.94	15			

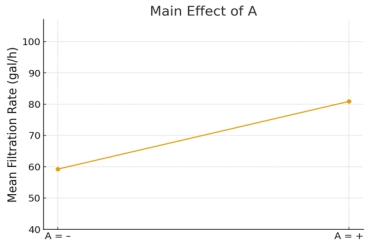
What if we include some other effects?

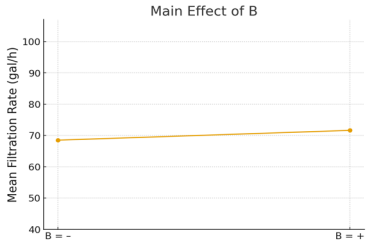
■ **TABLE 6.13**

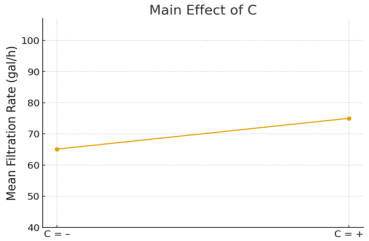
Analysis of Variance for the Pilot Plant Filtration Rate Experiment in *A*, *C*, and *D*

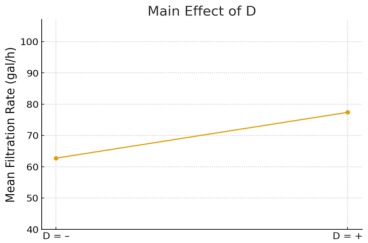
Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0	*P*-Value
A	1870.56	1	1870.56	83.36	<0.0001
C	390.06	1	390.06	17.38	<0.0001
D	855.56	1	855.56	38.13	<0.0001
AC	1314.06	1	1314.06	58.56	<0.0001
AD	1105.56	1	1105.56	49.27	<0.0001
CD	5.06	1	5.06	<1	
ACD	10.56	1	10.56	<1	
Error	179.52	8	22.44		
Total	5730.94	15			

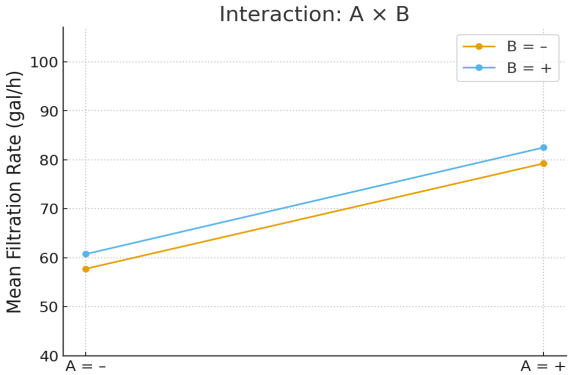
Main Effects & Two-Factor Interactions

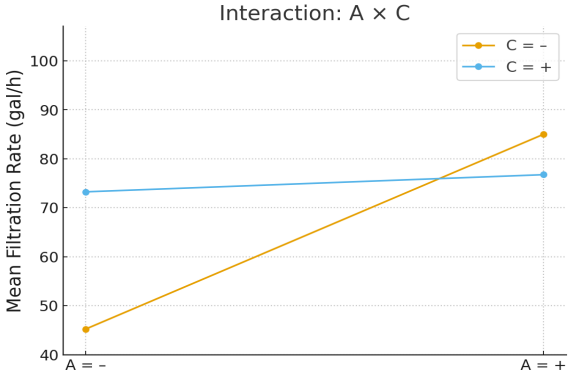


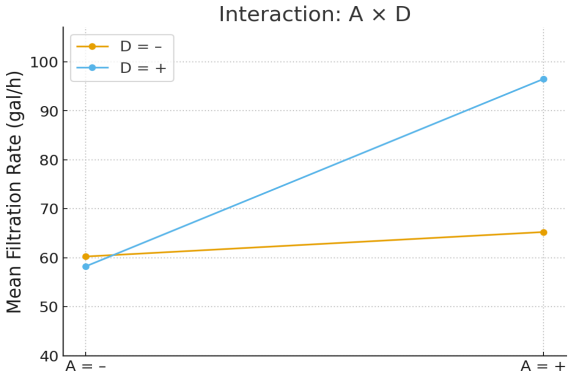


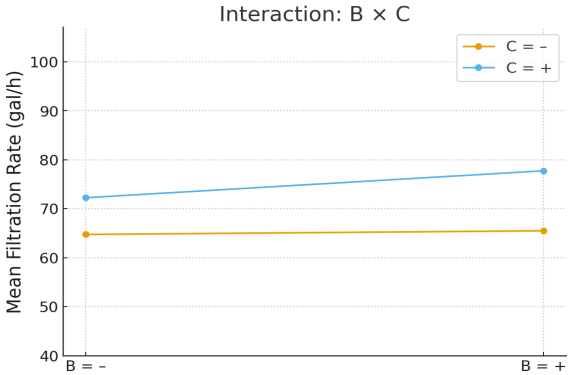


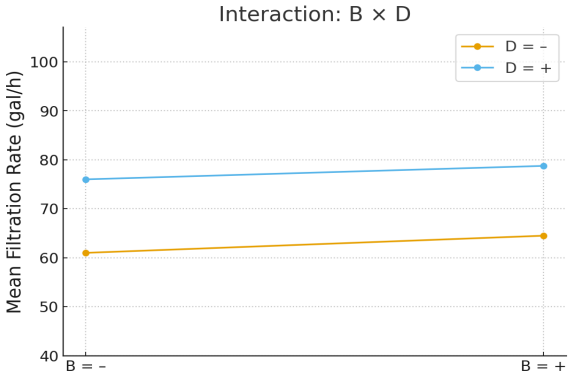


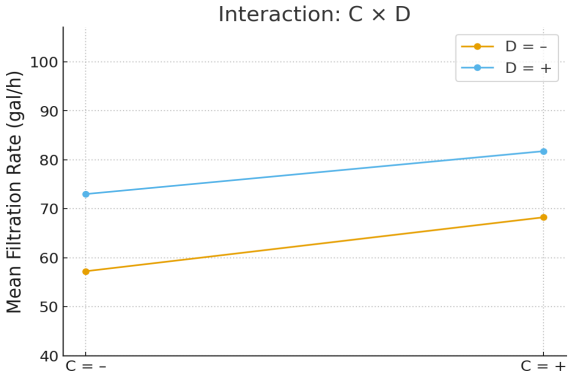








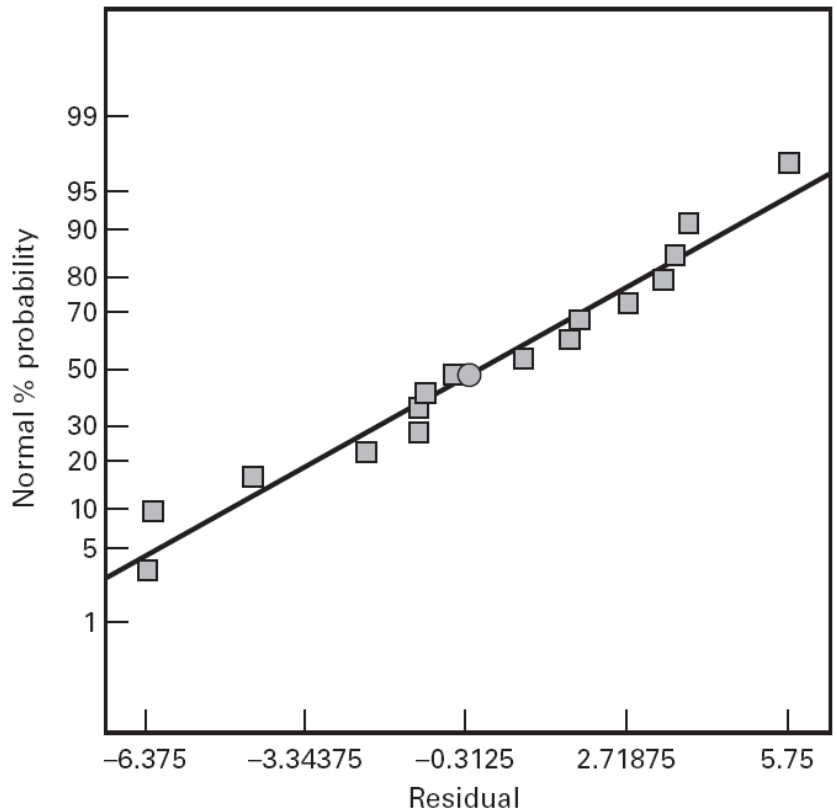




The Regression Model

$$\hat{y} = 70.06 + \left(\frac{21.625}{2}\right)x_1 + \left(\frac{9.875}{2}\right)x_3 + \left(\frac{14.625}{2}\right)x_4 - \left(\frac{18.125}{2}\right)x_1x_3 + \left(\frac{16.625}{2}\right)x_1x_4$$

	y	$\hat{y}$	$e = y - \hat{y}$
(1)	45	46.25	−1.25
a	71	69.38	1.63
b	48	46.25	1.75
ab	65	69.38	−4.38
c	68	74.25	−6.25
ac	60	61.13	−1.13
bc	80	74.25	5.75
abc	65	61.13	3.88
d	43	44.25	−1.25
ad	100	100.63	−0.63
bd	45	44.25	0.75
abd	104	100.63	3.38
cd	75	72.25	2.75
acd	86	92.38	−6.38
bcd	70	72.25	−2.25
$abcd$	96	92.38	3.63

Model Residuals are Satisfactory

■ **FIGURE 6.13**
Normal probability plot of residuals for Example 6.2